Srijon Mandal

Education

University of California, Berkeley | Bachelor's Degree – Data Science *Berkeley, CA · Aug 2026 – Present*

Activities & Societies: Regents and Chancellors' Scholarship Association.

Projects

RODAN: Portable AI Driver-Assistance Device | Sole Engineer *Remote · Jan 2023 – Present*

- Built a sub-$200 edge ADAS running a YOLO/MobileNet-SSD ensemble over LiDAR, thermal IR, and optical video at **>95% accuracy, <100ms latency** on Raspberry Pi + Coral Edge TPU.
- Designed a BLE offload pipeline for zero-cloud NPU retraining; **U.S. patent filed** (11039-10722-PR) and earned **U.S. Air Force Special Recognition for Innovation**.

COMPANO: IoT Companion Device for Mental Well-Being | Sole Engineer *Remote · Nov 2022 – Mar 2023*

- Prototyped a <$150 BOM edge companion achieving **95% emotion-detection accuracy at <200ms latency** via quantized TFLite CNNs on Raspberry Pi.
- Integrated hardware sensors, inference, and emergency triggers end-to-end; **provisional patent filed** (11039-10726-PR) to anchor commercialization.

Research & Engineering

Brain Computation Lab, UT El Paso | Research Engineer *Virtual · 2025 – Present*

- Accelerated tetrode neural-recording analysis for graduate researchers by shipping a MATLAB app implementing a novel spike-sorting algorithm.
- Replaced a greedy clustering step with a pairwise-affinity-matrix approach, with checkpointing and multiple grouping policies for high-dimensional spike data.
- Work contributes to a manuscript in preparation for *Cell Reports Methods*.

Alzheimer's Neural Dynamics Framework | Independent Researcher *Pleasanton, CA · 2024 – 2025*

- Architected a dual-model computational framework coupling the Kuramoto oscillator model with a Hopfield network to mechanistically link disrupted neural synchronization to memory decline.
- Presented at **ISMB/ECCB 2025** (largest international computational biology conference) and **SCCUR 2025**; submitted to Regeneron STS 2026.

ASDRP – Quantum ML for Medical Imaging | Researcher *Fremont, CA · 2023 – 2024*

- Built quantum CNNs and vision transformers in Qiskit for spinal-injury detection, benchmarking against classical baselines.
- Earlier rotation: predicted chemical structures from F19 NMR spectra using deep regression models.

Indoor Localization Framework – UCSC SIP | Research Intern *Santa Cruz, CA · Jun 2023 – Aug 2023*

- Built and deployed an iOS app using ARKit + RANSAC to extract 3D point clouds, fit geometric planes, and map them to floor plans for real-time indoor positioning – targeted at visually-impaired users.

TP53 Mutation Analysis for Precision Oncology | Independent Researcher *Pleasanton, CA · 2024 – 2025*

- Architected a hybrid statistical & deep-learning framework to classify TP53 mutations and surface candidate targeted therapies.
- Improved diagnostic accuracy over baseline classifiers and identified mutation-specific treatment pathways for personalized oncology.
- Presented at **JSHS 2025** (Junior Science and Humanities Symposium).

MSMAS: Schizophrenia Monitoring & Analysis System | Sole Developer *Pleasanton, CA · 2024*

- Shipped an AI framework that ingests multi-channel patient signals (speech, behavioral, clinical) and standardizes them into objective diagnostic outputs for clinical professionals.
- Replaced subjective single-modality assessment with reproducible scoring, reducing diagnostic variance across evaluators.

Thought-to-Text BMI – UCI x GATI | Researcher *Irvine, CA · Jun 2024 – Jul 2024*

- Designed a motion-controlled input device for disabled users using non-invasive brain-computer interface signals for tech interaction.
- **Published in *Youth Science Journal*** on thought-to-text processing via non-invasive BMIs.

Awards & Recognition

- **Alameda County Science & Engineering Fair:** 1st Place (2022, 2024, 2025); Grand Award – Engineering (2022).
- **California State Science & Engineering Fair:** 3rd Place (2022).
- **Math & CS:** USACO Gold; 2× AIME Qualifier (top 2.5%); SAT 1550.